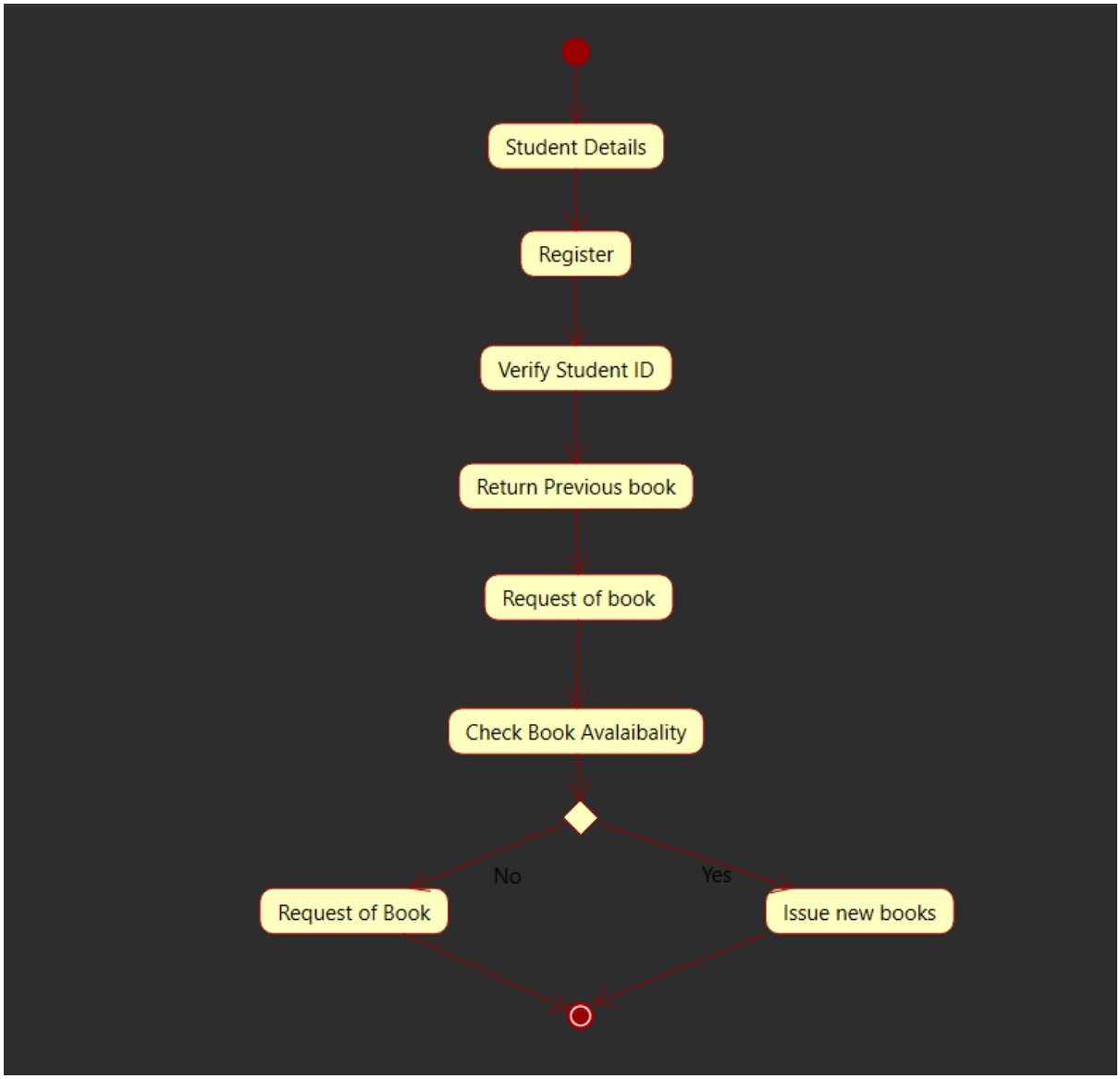
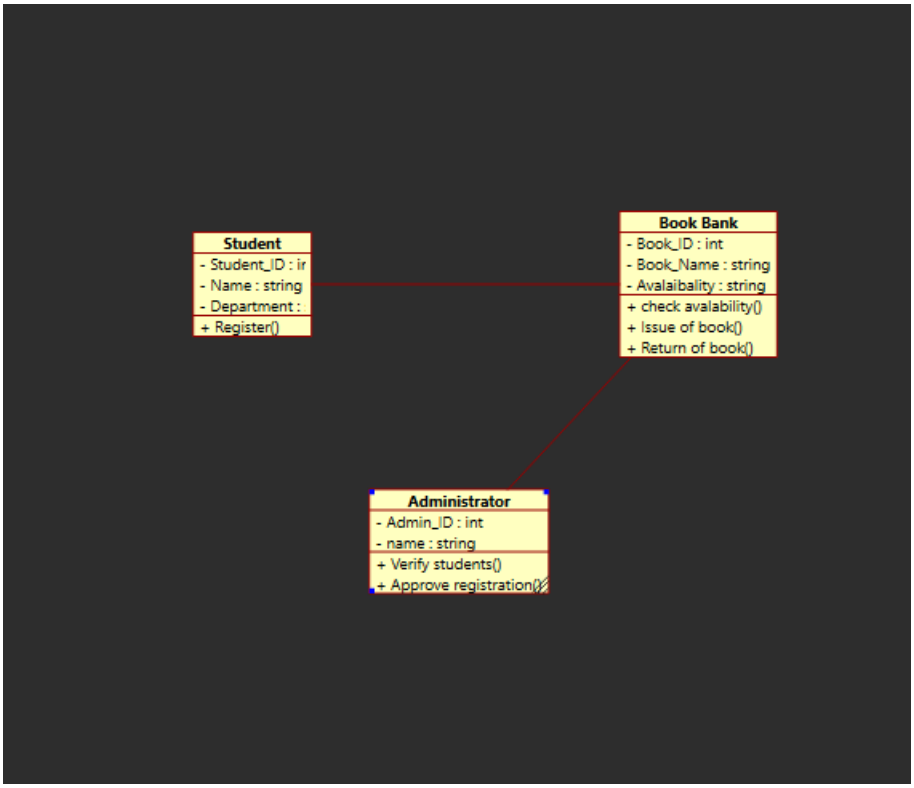
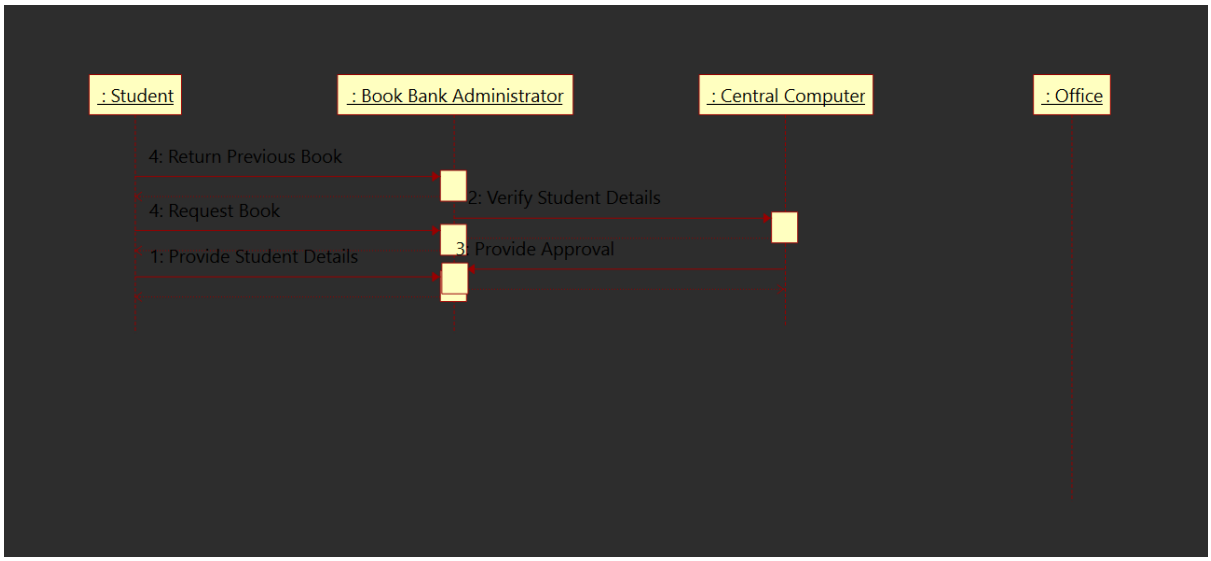
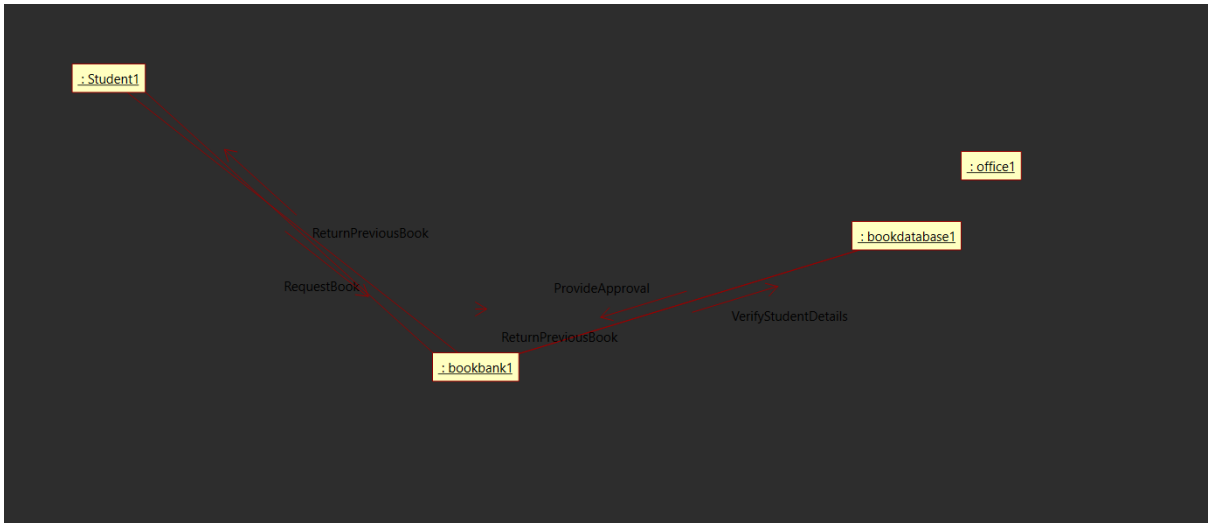
CSA1110- OBJECT ORIENTED ANALYSIS & DESIGN
EXP -2 BOOK BANK REGISTRATION SYSTEM
(UML DIAGRAMS)

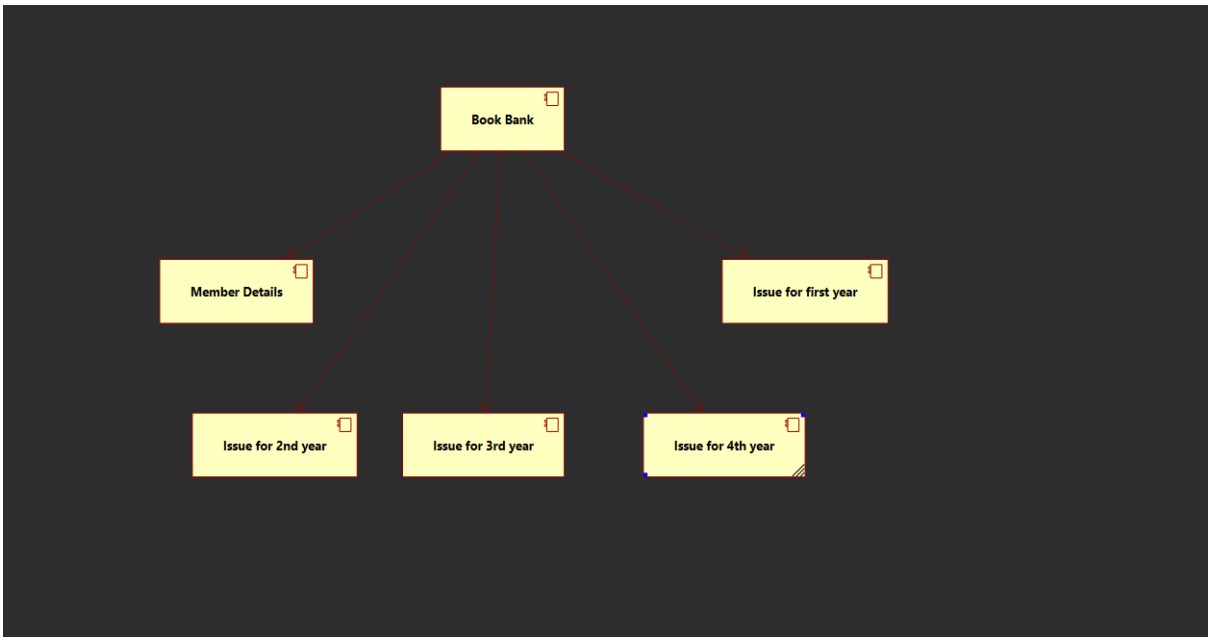


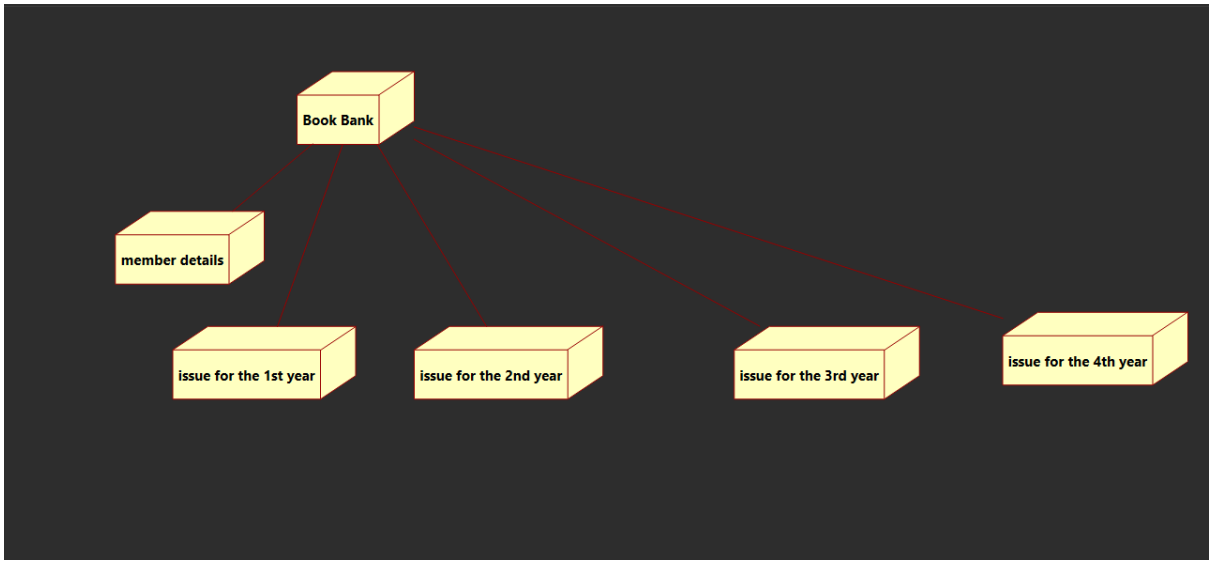

Student Details
Register
Verify Student ID
Return Previous book
Request of book
Check Book Avalaibality
No
Yes
Request of Book
Issue new books

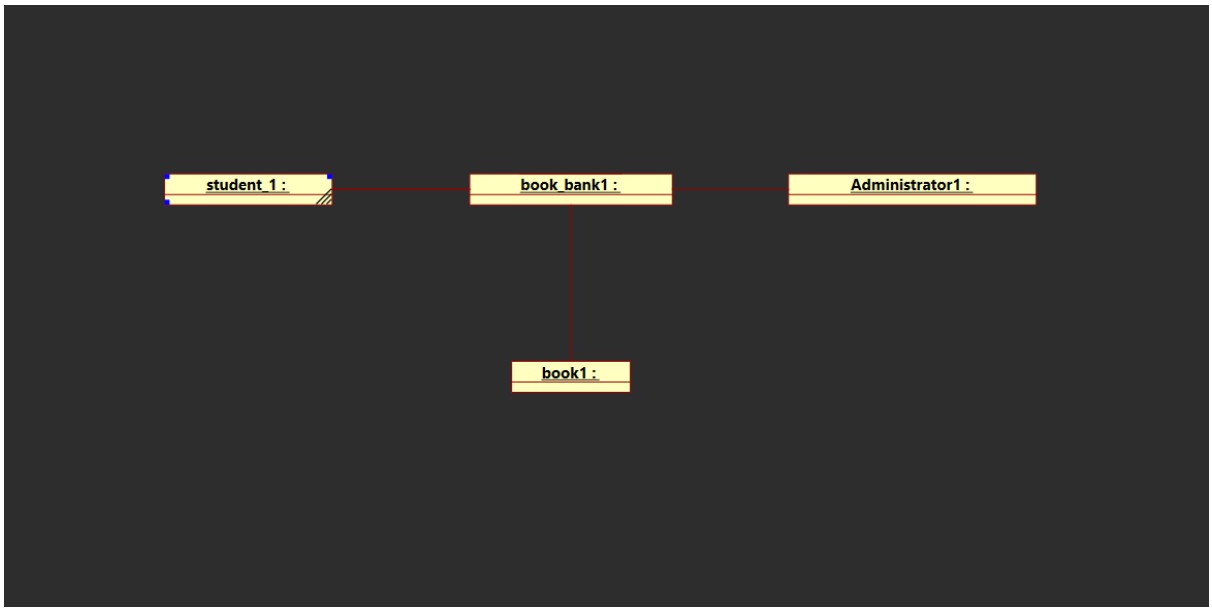

Student
- Student_ID : ir
- Name : string
- Department :
+ Register()
Book Bank
- Book_ID : int
- Book_Name : string
- Avalaibality : string
+ check avalability()
+ Issue of book()
+ Return of book()
Administrator
- Admin_ID : int
- name : string
+ Verify students()
+ Approve registration()

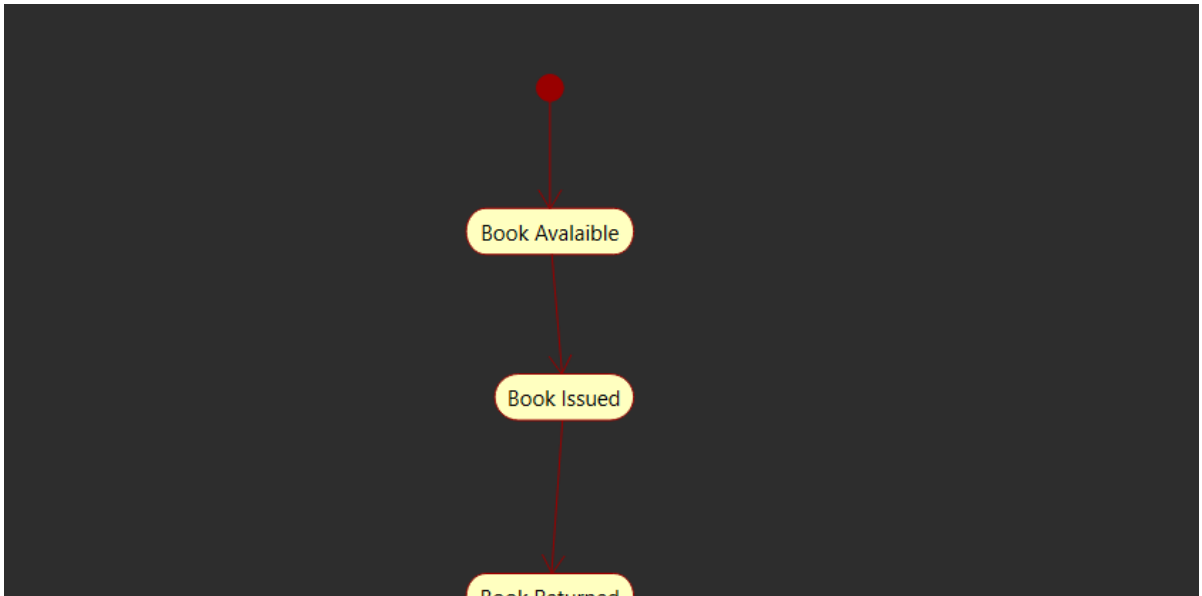

: Student
: Book Bank Administrator
: Central Computer
: Office
4: Return Previous Book
2: Verify Student Details
4: Request Book
3: Provide Approval
1: Provide Student Details


: Student1
: office1
: bookdatabase1
ReturnPreviousBook
RequestBook
ProvideApproval
VerifyStudentDetails
ReturnPreviousBook
: bookbank1


Book Bank
Member Details
Issue for first year
Issue for 2nd year
Issue for 3rd year
Issue for 4th year


Book Bank
member details
issue for the 1st year
issue for the 2nd year
issue for the 3rd year
issue for the 4th year


student_1 :
book_bank1 :
Administrator1 :
book1 :


Book Avalaible
Book Issued

CODE

```
from CENTRAL_COMPUTER import *

class ADMIN:

  """
  :version:
  :author:
  """

  """ ATTRIBUTES

  STUDENT_DETAILS  (private)

  BOOK_LIST  (private)

  """

  def verify_student_id_(self):
    """


    @return  :
    @author
    """
    pass

  def check_avalaibality_(self):
    """


    @return  :
    @author
    """
    pass

  def issue_books_(self):
    """


    @return  :
    @author
    """
    pass

  def order_new_books(self):
    """
```

```
    @return  :
    @author
    """
    pass

  def maintain_student_details(self):
    """


    @return  :
    @author
    """
    pass



class BOOK:

  """
  :version:
  :author:
  """

  """ ATTRIBUTES

  book_id  (private)

  book_name  (private)

  avalaibality  (private)

  """

  def check_avalaibality(self):
    """


    @return  :
    @author
    """
    pass

  def issue_book(self):
    """


    @return  :
    @author
```

```
    """
    pass


from BOOK import *

class CENTRAL_COMPUTER:

    """
    :version:
    :author:
    """

    """ ATTRIBUTES

    student_details  (private)

    """

    def verify_student_details(self):
        """


        @return  :
        @author
        """
        pass

    def provide_approval(self):
        """


        @return  :
        @author
        """
        pass


from ADMIN import *

class STUDENT:

    """
    :version:
    :author:
    """

    """ ATTRIBUTES
```

```
  student_if  (private)

  student_details  (private)

  """

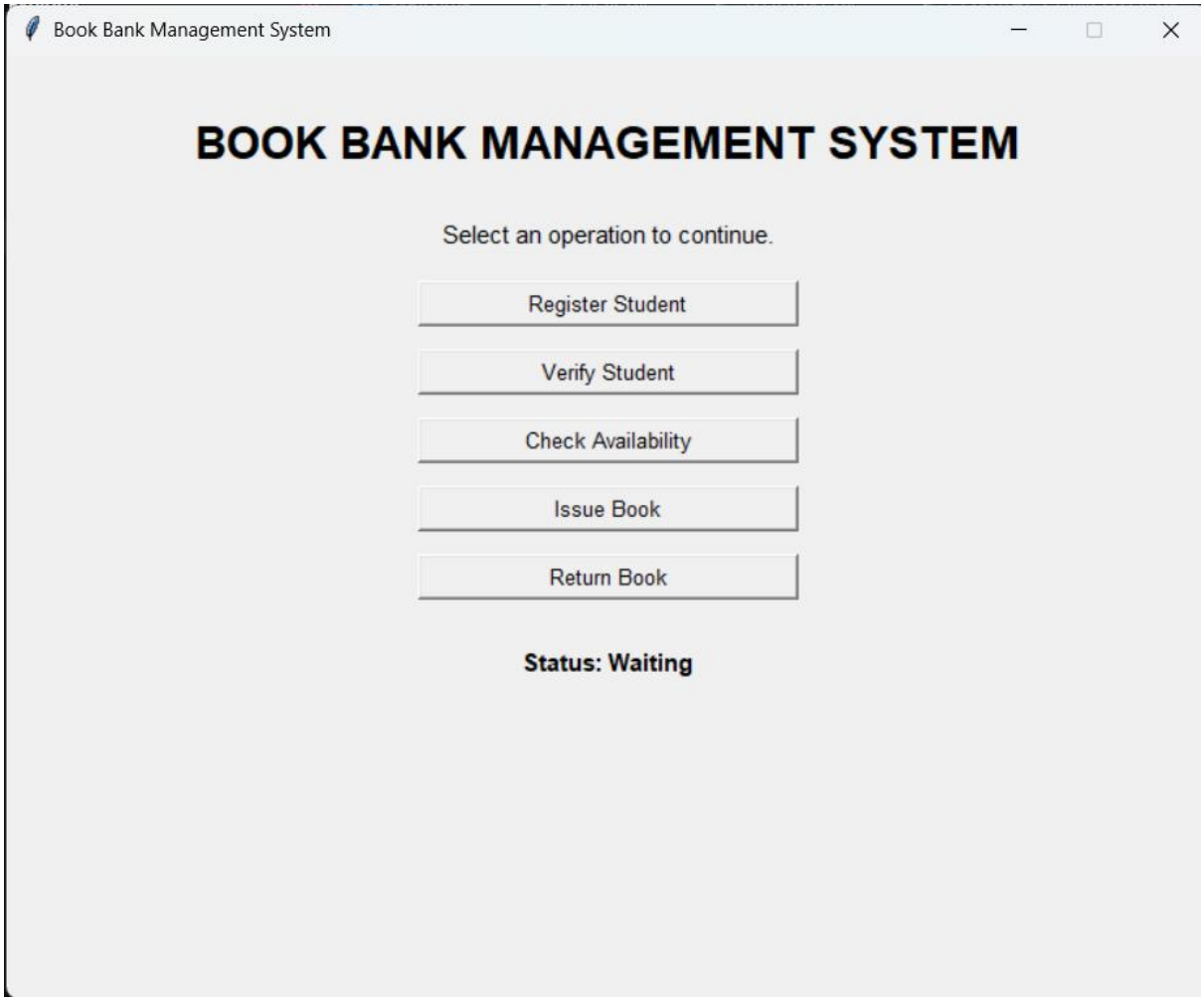
  def request_book(self):
    """

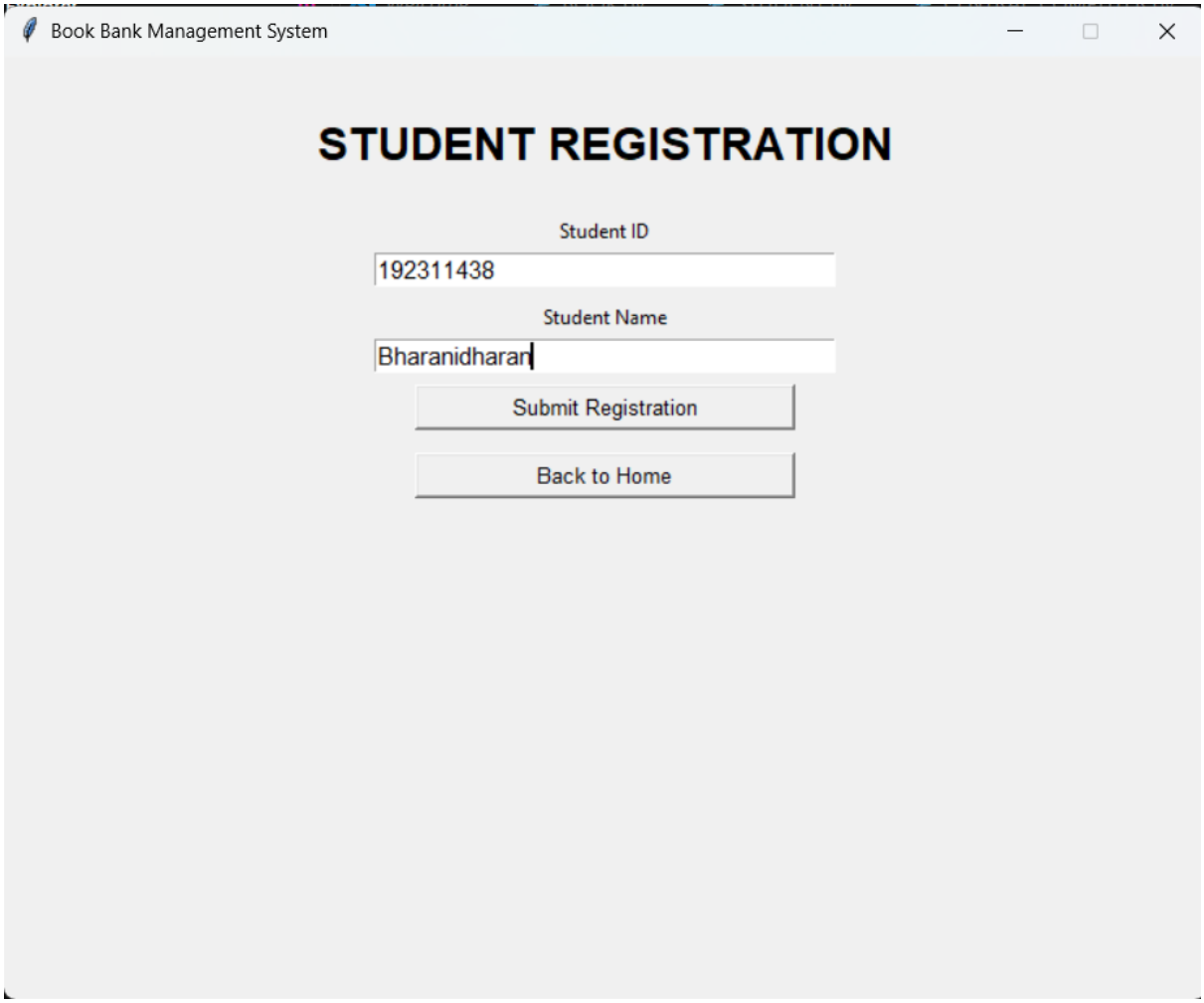

    @return  :
    @author
    """
    pass

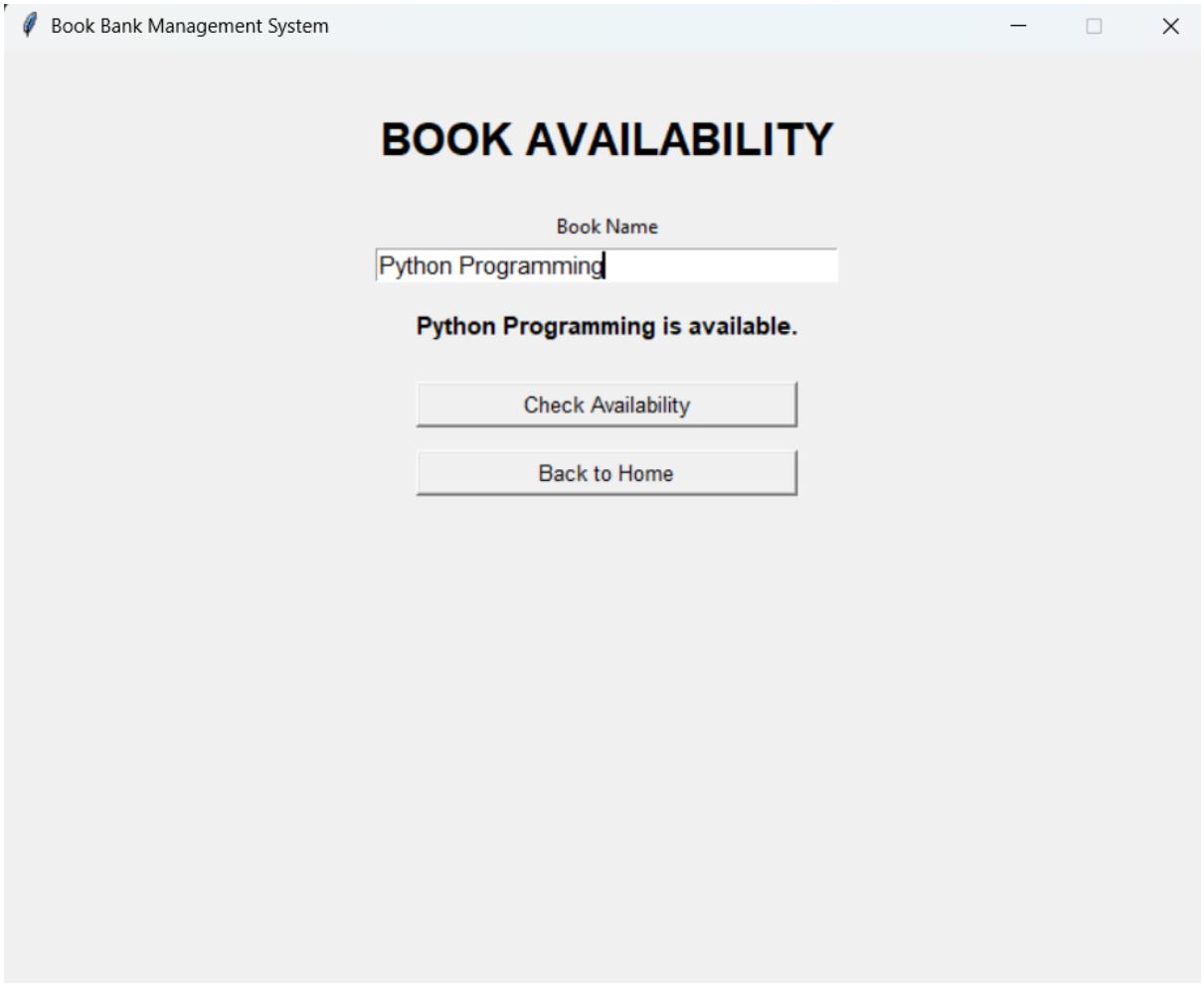
  def return_book(self):
    """

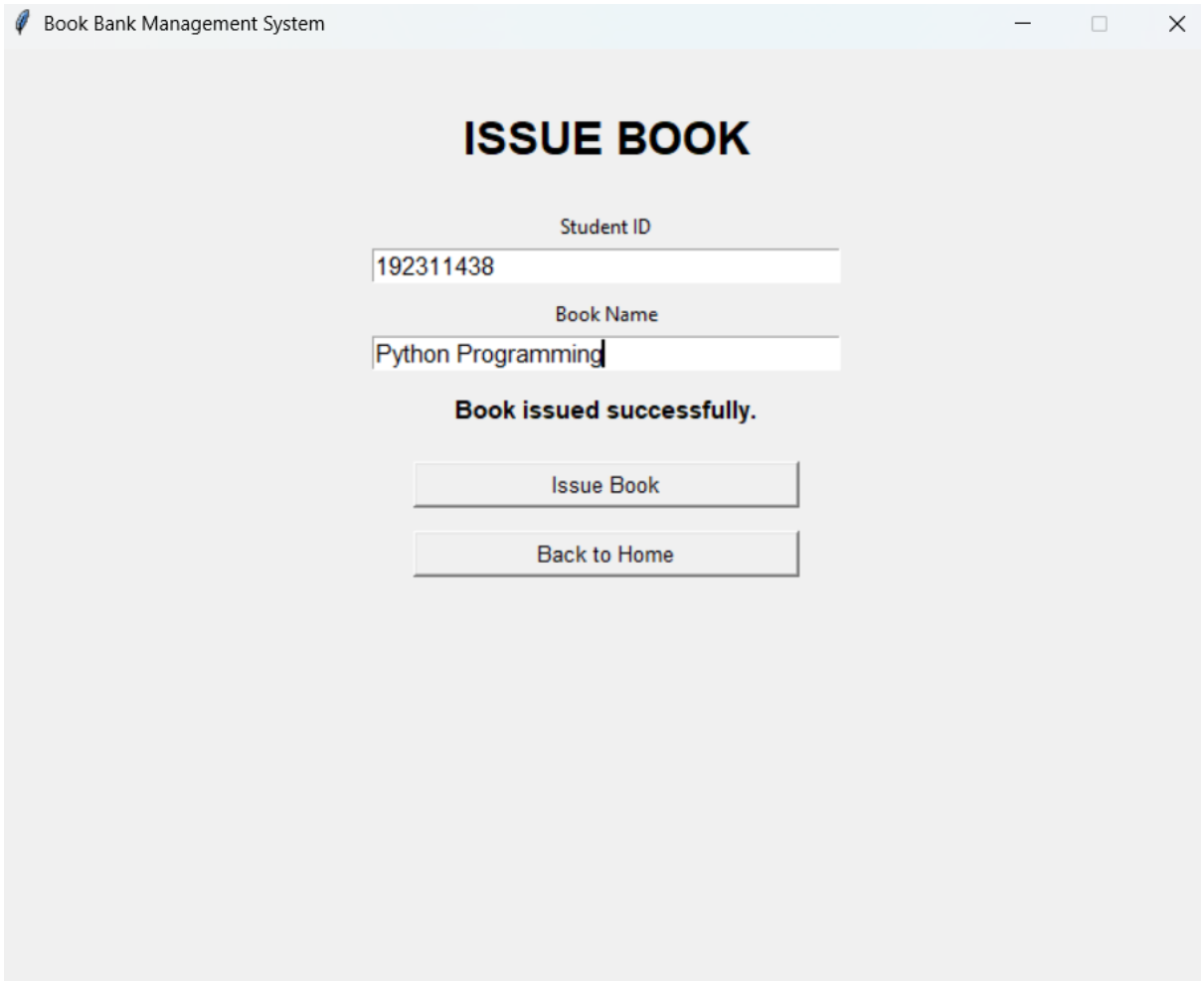

    @return  :
    @author
    """
    pass
```

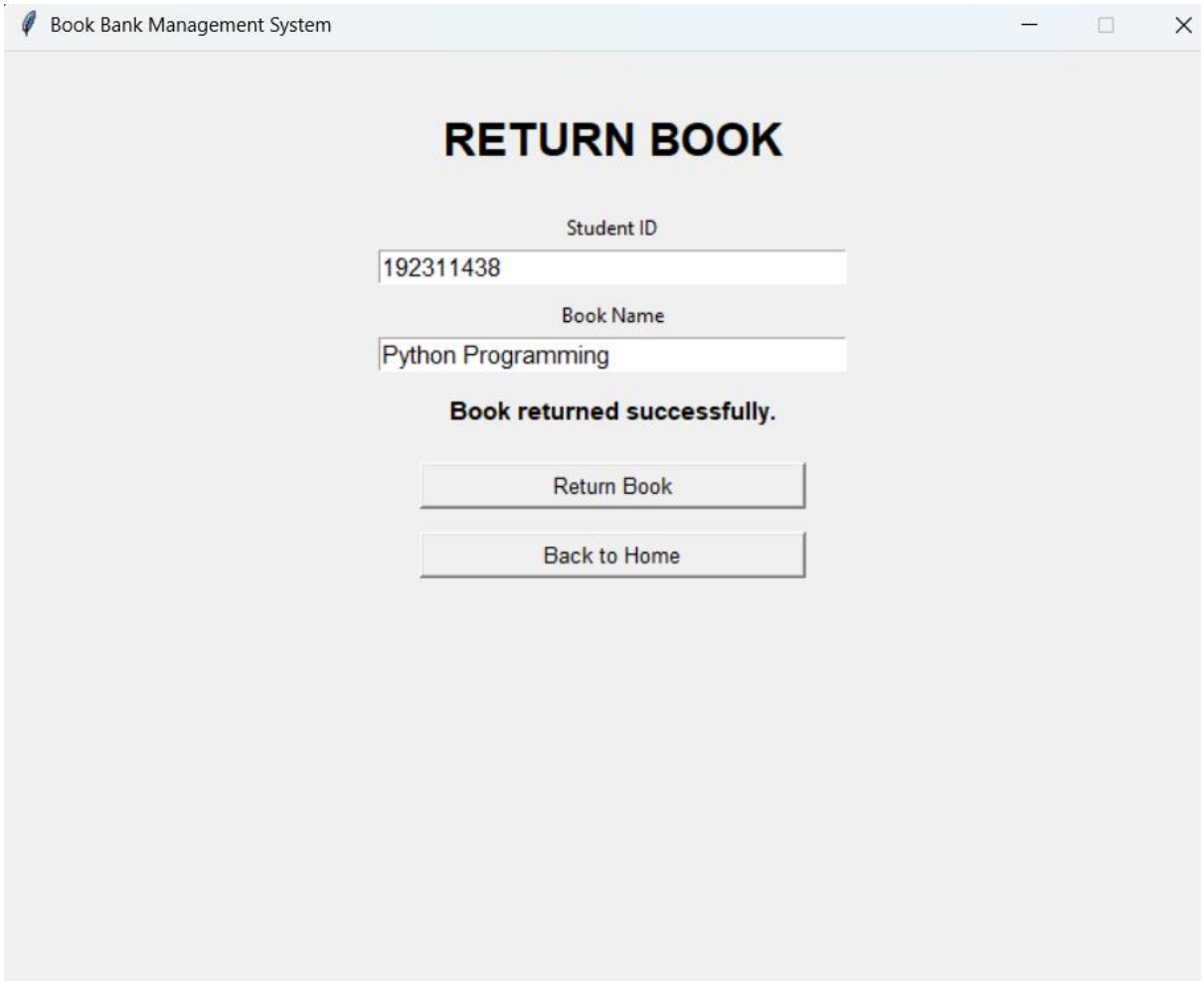
OUTPUT


Book Bank Management System
BOOK BANK MANAGEMENT SYSTEM
Select an operation to continue.
Register Student
Verify Student
Check Availability
Issue Book
Return Book
Status: Waiting


Book Bank Management System
STUDENT REGISTRATION
Student ID
192311438
Student Name
Bharanidharan
Submit Registration
Back to Home


Book Bank Management System
BOOK AVAILABILITY
Book Name
Python Programming
Python Programming is available.
Check Availability
Back to Home


Book Bank Management System
ISSUE BOOK
Student ID
192311438
Book Name
Python Programming
Book issued successfully.
Issue Book
Back to Home


Book Bank Management System
RETURN BOOK
Student ID
192311438
Book Name
Python Programming
Book returned successfully.
Return Book
Back to Home